

Comparative Study of Layer-Reduction Approaches

Sumit Jha

October 2026

Abstract

We implemented fifteen layer-reduction ideas for GPT-2 (124M) and SmolLM-135M and evaluated each on `wikitext-2-raw-v1` with a fixed perplexity protocol, repeated over three seeds with per-fit seeding. Eight produced a usable signal; seven failed by orders of magnitude or were skipped for stated reasons.

Three results reorganise the study away from its original hypothesis. First, the approach that motivated this project, a fitted ternary low-rank bridge, *loses* to plain deletion while achieving identical parameter counts. Second, deleting the lowest-influence block and repairing it with pruned distillation works on both models, recovering 35 to 54 percent of the perplexity that deletion costs. Third, LayerDrop fine-tuning is the strongest single method on SmolLM-135M and is the only configuration in the whole study that lands below its own unpruned baseline, which makes the best method model-dependent rather than universal.

This document walks through all fifteen, gives the mathematics of the ones that ran, and states the failure modes of the ones that did not.

1 Setup used for every approach

Models: GPT-2 small (124,439,808 parameters, 12 blocks) and SmolLM-135M (134,515,008 parameters, 30 blocks), loaded with `use_cache=False`. Data: `Salesforce/wikitext`, config `wikitext-2-raw-v1`. Evaluation:

$$\text{PPL} = \exp\left(\frac{1}{N} \sum_{i=1}^N -\log p_{\theta}(x_i \mid x_{<i})\right)$$

computed over non-overlapping windows, identical token budget for every approach.

Seeding, and why it changed the conclusions. Every stochastic fit is driven by an explicit per-fit `torch.Generator`, not by a single global seed. An earlier version of this study used a global seed and produced figures that could not be reproduced: the same configuration yielded 47.67/27.04 when re-executed alone and 47.95/27.67 when reached by running the notebook end to end. Those figures appeared in an earlier draft of this document and have been removed. All numbers below are the seeded ones.

A caution on GPT-2 and SmolLM internals. They are not interchangeable. GPT-2 attention builds its causal mask from `layer_idx`, so blocks must be re-indexed after any removal or the model silently attends with the wrong mask. GPT-2 also stores projections as `Conv1D` with weight shape `[in, out]`, whereas SmolLM’s linear projections are `[out, in]`. Code that treats both as `nn.Linear` corrupts one of them.

2 BI ranking, common to most approaches

For decoder block l we record

$$\text{BI}_l = 1 - \mathbb{E}_t[\cos(h_t^{(l)}, h_t^{(l+1)})]$$

averaged over the batch and sequence. Low BI_l means the block is near the identity $h^{(l+1)} \approx h^{(l)}$, so it is the natural first candidate for deletion, merging, or aggressive quantization. GPT-2’s most redundant blocks, ascending, are 4, 1, 2, 3, 5, 6; SmolLM’s are 2, 27, 5, 4, 3.

3 What we ran and how each came out

3.1 01 - Fitted-ternary TerGRASP bridge (RUNS, but LOSES)

Idea. For each redundant block L , collect $X = h_{in}$, $Y = h_{out}$ on calibration text, fit

$$\min_{U,V} \|XUV - \Delta\|_F^2, \quad \Delta = Y - X,$$

then AbsMean-ternarize U, V :

$$\gamma = \frac{1}{mn} \|W\|_1, \quad W_q = \gamma \text{clip}(\text{round}(W/\gamma), -1, 1), \quad W_q \in \{-\gamma, 0, \gamma\}.$$

The block is replaced by $g(x) = x + xU_qV_q$. **Original hypothesis.** At a low-BI position Δ is small and lives on a low-dimensional manifold, so a rank-32 factor recovers most of it. **Result.** The hypothesis is false at this parameter budget. GPT-2 $k=1$: 58.67 ± 9.53 against 45.76 for plain deletion. SmolLM $k=1$: 30.77 ± 3.54 against 19.93. Worse at every k . **Why it loses.** It is strictly dominated. A rank-32 bridge is 49,152 weights replacing a 7.1M block, so bridge and deletion yield *identical* parameter counts (117,351,936 for both on GPT-2 at $k=1$, against 124,439,808 baseline) and indistinguishable latency (26.54 against 26.56 ms). The bridge therefore deletes exactly the same weights as deletion and then adds a perturbation on top. It can only lose. An earlier draft of this document reported this approach as matching or beating hard removal (45.11 and 21.71). Those were the irreproducible global-RNG figures.

3.2 06 - ShortGPT BI baseline (WORKS, the control)

Idea. Remove the k lowest-BI blocks; the control removes k random ones drawn *disjointly* from the BI set, so the null can never receive the block we selected. **Result.** Disjointness matters. GPT-2: 45.76 vs 1453.93 ± 2441.72 at $k=1$, 47.95 vs 4351.09 ± 203.58 at $k=2$, 121.81 vs 2814.29 ± 2378.60 at $k=3$. SmolLM: 19.93 vs 20.16 ± 1.94 at $k=1$, 26.13 vs 22417.94 ± 19512.02 at $k=2$, 29.91 vs 269.08 ± 287.45 at $k=3$. **An honest qualification.** At SmolLM $k=1$ the two are within noise. The claim that BI beats random *by orders of magnitude at every point* is false and we measured it. What survives is a reliability claim: BI is never worse at any $(model, k)$, and the advantage is concentrated in the worst case over seeds. Random removal on GPT-2 has a standard deviation exceeding its own mean, which is exactly the property that makes a configuration undeployable.

3.3 03 - Pruned distillation repair (WORKS, now the headline)

Idea. LLM-Pruner showed depth-pruned models need distillation, so comparing two unrepaired operators only answers which does less damage. We repair both. The loss is

$$\mathcal{L} = \alpha \text{CE}(y, \hat{y}_S) + (1 - \alpha) T^2 \text{KL}(\text{softmax}(\hat{y}_S/T) \parallel \text{softmax}(\hat{y}_T/T)),$$

S the pruned student, T the frozen original. The T^2 factor is not cosmetic: softening a 50k-way distribution by T scales the gradient by $1/T^2$, so omitting it silently downweights distillation until the run is merely fine-tuning. **Result.** GPT-2 $k=1$: $45.76 \rightarrow 41.68 \pm 0.37$, recovering 54.3%. GPT-2 $k=2$: $47.95 \rightarrow 44.06 \pm 0.17$, 40.2%. SmolLM $k=1$: $19.93 \rightarrow 18.67 \pm 0.07$, 34.9%. SmolLM $k=2$: $26.13 \rightarrow 22.14 \pm 0.09$, 39.4%. Healed beat pruned in 12 of 12 runs. **Correction to an earlier draft.** That draft claimed the KL loss was “essentially zero” at a 60-step budget, so repair was a no-op. That was a budget artefact, not a property of distillation: at 400 steps the mean initial KL is 134.48 falling to 34.81, which is a very large distillation signal. We now assert the KL is non-trivial rather than assuming it.

3.4 11 - LayerDrop (WORKS, and better than expected on SmolLM)

Idea. Fine-tune with structured layer dropout at $p = 0.15$ for 500 steps, so the surviving network adapts to the removal. **Result.** GPT-2 $k=1$: 51.74 ± 0.13 , $k=2$: 57.53 ± 0.28 , both worse than plain deletion. SmolLM $k=1$: 15.54 ± 0.01 against a 16.39 baseline, $k=2$: 19.62 ± 0.16 . **Why this matters.** On SmolLM, LayerDrop lands *below* the unpruned baseline: deleting a block and fine-tuning made the model better, with a standard deviation of 0.01. It also beats distillation repair on the same model (15.54 against 18.67). It does not generalise, since on GPT-2 it is far worse than either deletion or repair. The best method is therefore model-dependent, and recommending one universally would overstate the evidence.

3.5 10 - Learned L0 gates (RUNS, never beats BI)

Idea. Train a per-block gate with an L_0 penalty, then delete the lowest-gate blocks. **Result.** GPT-2 $k=1$: 44.25, $k=2$: 59.14. SmolLM $k=1$: 18.95, $k=2$: 24.48. Never better than the BI ranking it is meant to improve on. **The interesting failure.** Overlap between the learned gate ordering and the BI ordering is exactly 0.00 on GPT-2. The gates identify a completely different set of blocks, most often keeping block 0 and removing blocks 1, 2 and 5, yet arrive at a similar perplexity. Gates are a more expensive way of rediscovering that most blocks are not equally important.

3.6 12 - Attention-head pruning by magnitude (WORKS, clean negative)

Idea. Prune the least important attention heads by magnitude. **Result.** GPT-2: $38.25 \rightarrow 42.85$ at 12.5% pruned, $\rightarrow 2033.08$ at 25%. SmolLM: $16.36 \rightarrow 46.92$ at 12.5%, $\rightarrow 102.70$ at 25%. Affordable at 12.5% on GPT-2, catastrophic past that, and severe immediately on SmolLM.

3.7 07 - SVD low-rank truncation (RUNS on SmolLM)

Idea. For each projection W in the redundant blocks, $W \approx U_r S_r V_r^\top$ with rank r . **Maths.** By Eckart–Young the reconstruction error is the tail $\sum_{i>r} \sigma_i^2$, small when W is numerically low-rank. **Result.** SmolLM is smooth in r as expected. An earlier draft described the GPT-2 rows as “a dtype-guard artifact”; that was a bug we had not isolated, and it should not have been described as noise. The mechanism analysis in the main report supersedes this approach anyway, because it shows local rank-truncation quality does not track end-task perplexity.

3.8 15 - Quantized no-op bypass (RUNS, supporting rule)

Idea. Quantize every weight inside the lowest-BI blocks to $b \in \{8, 4, 2\}$ bits, AbsMean symmetric, and measure PPL. **Maths.** If $F_L(x) = x + \Delta(x)$ with $\|\Delta\|$ small, the layer’s output is insensitive to bounded weight perturbations $\|W - W_q\|_\infty \leq \gamma$, so the induced output change is $\leq \|\Delta W\| \|x\|$, tiny next to $\|x\|$. **Result.** SmolLM L5: 17.76, 17.77, 17.76 across 8, 4 and 2 bits, i.e. essentially flat. SmolLM L2: 19.11, 19.07, 19.18. **Rule.** BI answers both questions: can this block be deleted, and can its precision be crushed?

3.9 13 - Early exit (DELIBERATELY SKIPPED)

Not attempted, and we think that is correct rather than convenient. Early exiting changes depth *dynamically* per input rather than removing blocks from the model, so it is not a layer-reduction method. It also requires trained exit heads, which is a training project of its own.

4 What failed, with the exact failure mode

4.1 02 - Affine bridge across the deleted gap

We fit one ridge affine map $Y \approx AX + b$ from the input of the first removed block to the output of the last and inserted it as a single `nn.Linear`. **Why it fails.** The true map $F_{last} \circ \dots \circ F_{first}$ is not affine; it is a composition of LayerNorms, matmuls and nonlinearities, so one line cannot approximate it. PPL reached 4.5×10^3 on GPT-2 and 1.9×10^8 on SmolLM. **This was the motivation for fitting per layer, which 01 then showed is still not enough.**

4.2 08 - Parameter soup of adjacent blocks

$\theta_{merge} = \alpha\theta_L + (1 - \alpha)\theta_{L+1}$. **Why it fails.** Weight-space averaging across a neuron-permutation boundary is meaningless; GPT-2 $k=1$ gave 103 to 252 PPL against a baseline of 38.

4.3 14 - Cross-layer weight tying

Assigning `blocks[l+d] = blocks[l]` shares one set of weights between two positions. **Why it fails.** Depth is not redundant in weight space; tying at any period d exploded PPL to 10^3 to 10^6 times baseline.

4.4 04 - Permutation-aligned averaging

Capture per-neuron activations of both adjacent blocks, run Hungarian matching to align neurons, then average. **Why it fell back.** The Hungarian step was never hardened up and the fallback was the naive average, so the result equals 08 and inherits its failure.

4.5 09 - KD to a shallow student

Wired correctly, but a shallow student changes capacity rather than depth, so it answers a different question from the rest of this study. Reported as partial at the budget available.

4.6 05 - LayerDrop→BI→SVD→KD pipeline

Executed but produced flat rows because each stage was a scaffold rather than a completed implementation. Retained in `archived/` as incomplete. Its constituents are covered properly by 11, 06, 07 and 03 above.

5 How the surviving methods compare

	GPT-2 $k=1$	GPT-2 $k=2$	SmolLM $k=1$	SmolLM $k=2$
baseline (unpruned)	38.25	38.25	16.39	16.39
06 BI removal	45.76	47.95	19.93	26.13
01 fitted bridge	58.67	86.49	30.77	41.77
03 BI + distillation	41.68	44.06	18.67	22.14
11 LayerDrop	51.74	57.53	15.54	19.62

Approach	Best PPL	Params effect	Training?	Note
11 LayerDrop	SmolLM 15.54, below baseline	block removed	500 steps	best on SmolLM
03 BI + distillation	GPT-2 41.68	block removed	400 steps	best on GPT-2
06 BI removal	GPT-2 45.76	block removed	none	the control
01 fitted bridge	GPT-2 58.67	<i>identical</i> to removal	80 steps	dominated
10 learned gates	GPT-2 44.25	block removed	300 steps	disjoint from BI
12 head pruning	GPT-2 42.85 at 12.5%	width	none	collapses at 25%
07 SVD truncation	SmolLM smooth in r	width	none	superseded by mech.
15 quantized bypass	SmolLM 17.76	$\sim 16\times$ on block	none	rule of thumb
02/08/14/04	PPL 10^3 - $10^8\times$	irrelevant	none	negative results

6 Why the headline changed

An earlier version of this study recommended 01, the fitted ternary bridge, on the grounds that it was the only approach that simultaneously kept graph depth, shrank the replaced block, and held perplexity at baseline. With per-fit seeding and error bars, all three of those claims failed: the bridge does not hold perplexity, and it does not shrink anything, because at rank 32 its parameter count is identical to plain deletion. Keeping graph depth turned out to be worth nothing if the block you kept there is a worse approximation than nothing at all.

What survives instead is a recommendation that is honest about its own scope: delete the lowest-BI block and repair it with distillation, which is the best method on GPT-2, and note that LayerDrop is better on SmolLM-135M and is the only method here that improves on the unpruned baseline. The most useful finding came from an approach that was not the point of the project.

Reproducing this

`assignment2_tpu.ipynb` regenerates the primary comparison, the repair experiment, the mechanism diagnostic and the ablations end to end on Colab with a single **SMALL** flag. Source and per-approach results are at github.com/sumitjhaa/layer-reduction-approaches.